1 S-R-A LOOP

S-R-A loop is repeating process where:

1. Seasoning,
2. Reasoning and
3. Acting

is involved during the procedure of controlling the robot. This is the most important part of software in robotics. Remember the `autonomous` control is `ability to perform intended tasks based on current state and sensing, without human intervention`.

1.1 Tasks:

1. Write a program to drive the robot around the class and avoid the obstacles.
2. Using the S-R-A loop technique you should write the program in particular order:
    1. Check the sensor. IF the bumper ...
    2. ... Is pressed the robot has to stop/go back/turn.
    3. ... Is not pressed the robot can drive forward.

```
1   const int BUMPER_PIN = A0;
2   [+]void robotForward() { ...
3   [+]void robotStop() { ...
4   [+]void robotLeft() { ...
5   [+]void robotRight() { ...
6   [+]void robotBackward() { ...
7   [+]void setup() { ...
8      void loop() {
9        bool bumperIsPressed = digitalRead(BUMPER_PIN);
10      if ( bumperIsPressed ){
11      robotStop();
12      }else{
13      robotForward();
14      }
15     }
```

1.2 Questions:

1. What for `S-A-R loop` stands for?
2. Mark all three basic S-A-R processes in previous code example.
3. Can `line 4` of the previous code example be written outside of `loop()` function?

1.3 Summary:

1.3.1 <++>

<++>

1.4 Issues:

1.4.1 <++>

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